Simulating Kicked Floquet Ising System

Qiskit Fallfest •Track 3

-Schrodinger's Coders

Floquet evolution operator

Simulation Details

- **Floquet Ising Model:-**

$$H(t) = J\sum_{i=1}^{L-1} Z_i Z_{i+1} + h_z \sum_{i=1}^{L} Z_i + \frac{\theta}{2}\sum_{n=-\infty}^{\infty} \delta(t - nT) \sum_{i=1}^{L} X_i$$

where L is the number of spins, J_z is the nearest-neighbor Ising interaction strength, θ is the transverse field, and h_z is the longitudinal field.

- **Floquet evolution operator:-**

$$U_F = \exp\left(-i\frac{\theta}{2}\sum_{i=1}^{L} X_i\right) \exp\left[-iT\left(J\sum_{i=1}^{L-1} Z_i Z_{i+1} + h_z \sum_{i=1}^{L} Z_i\right)\right]$$

- **Imbalance:-**

$$\hat{\mathcal{I}} = \frac{1}{L}\sum_{i=0}^{L-1} (-1)^i Z_i$$

Initial state → $|0101\cdots01\rangle$ **or** $|\uparrow\downarrow\uparrow\downarrow\cdots\uparrow\downarrow\rangle$

I(t) ranges from +1 to -1.

*No Trotter error ZZ and Z terms commute and X term is responsible for periodic kick

Circuit

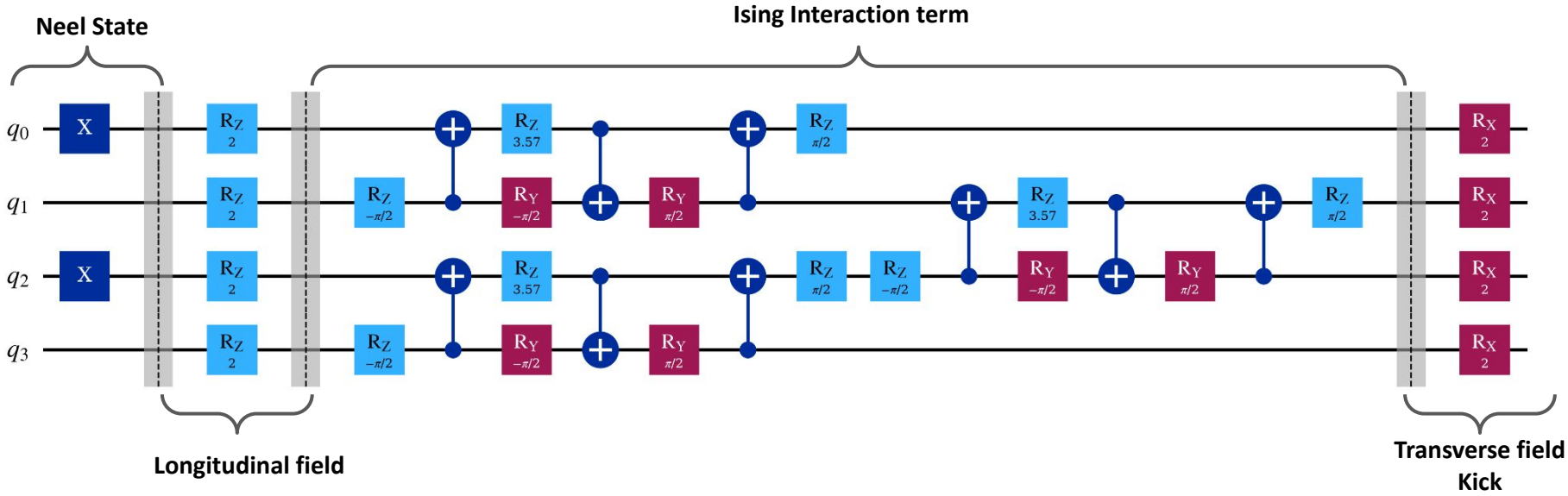
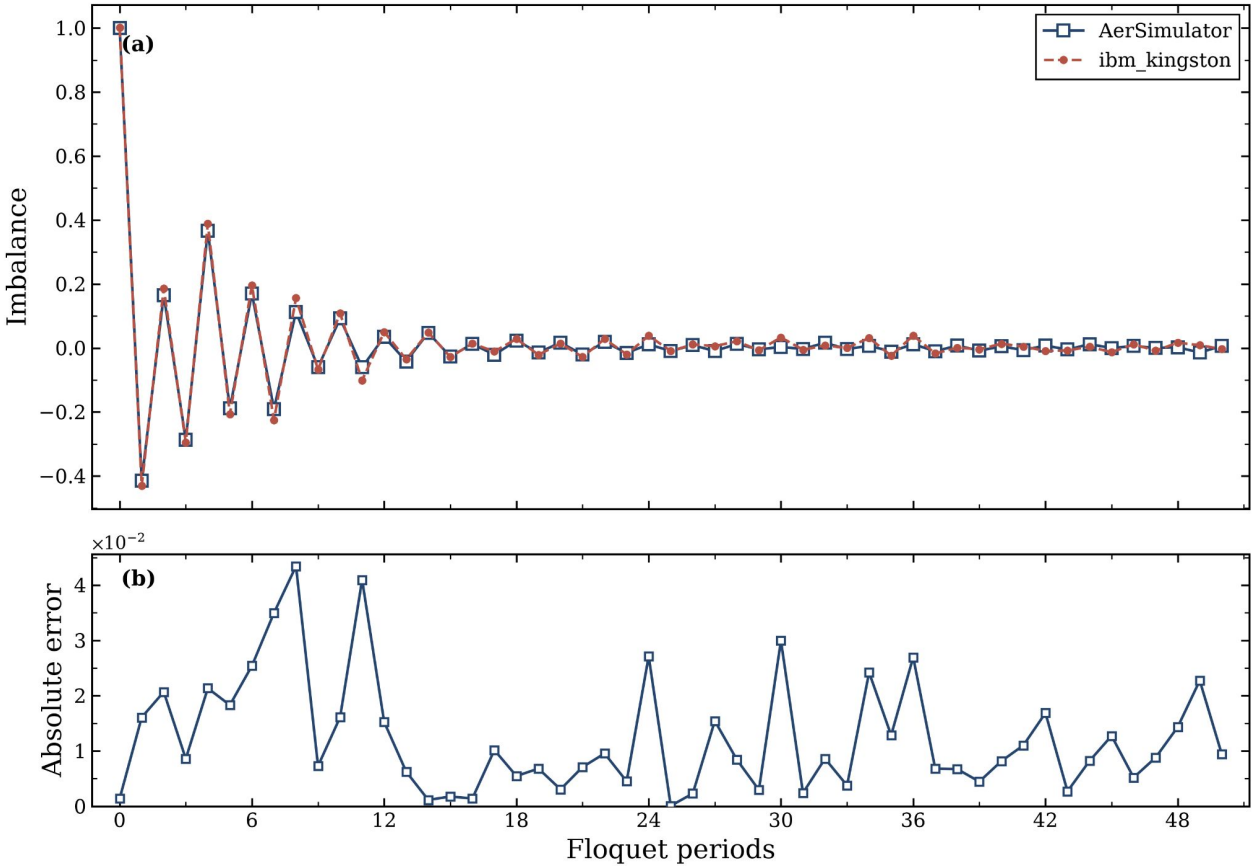


Figure.(Qiskit) Circuit for Kicked Floquet Ising model (N=4 one Floquet period)

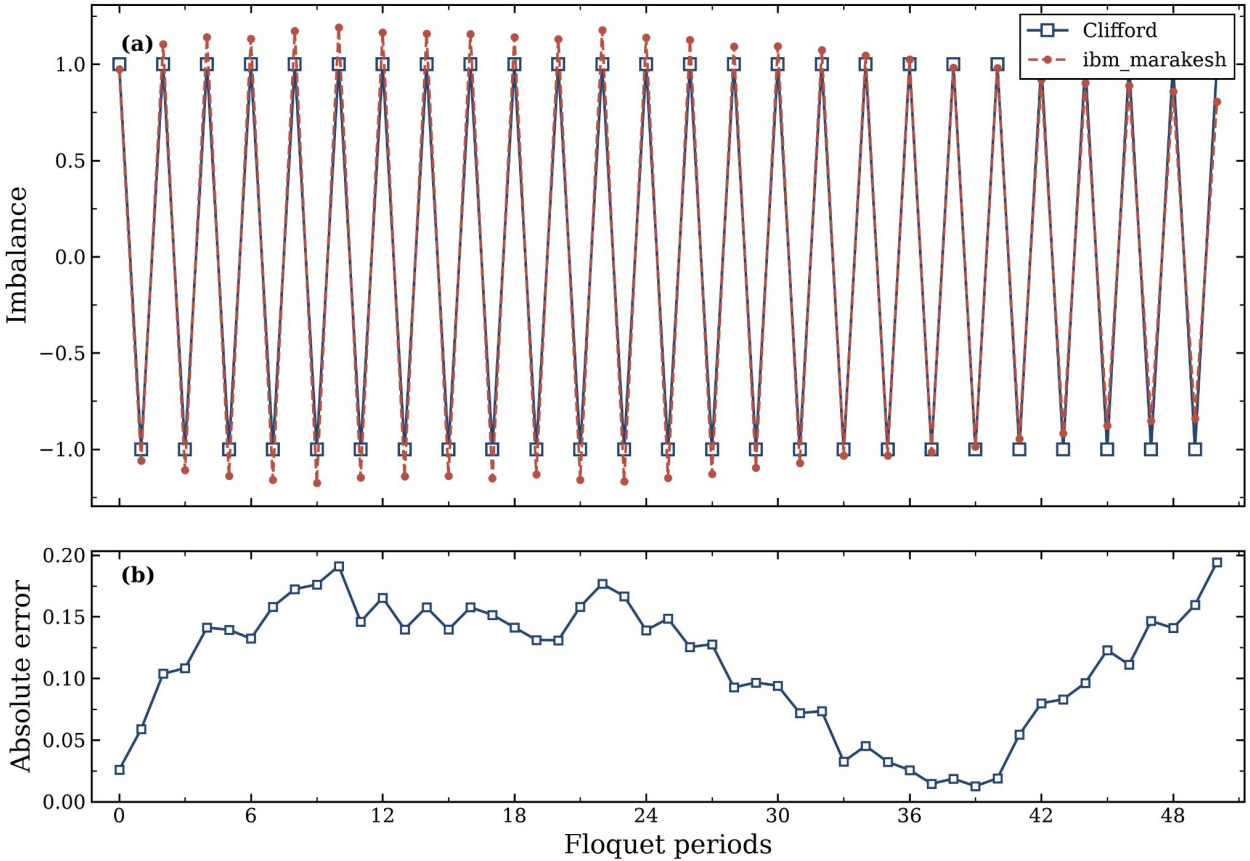
Results(L=20)



Parameters	Values
L(num_qubits)	20
Jz	1
hx (θ/2)	1
hz	1
Error Mitigation	ZNE (1, 1.2, 1.4) + DD+ MM + Twirling
Optimisation level	3

Figure.(expm vs QPU) Imbalance vs time for N=20 Kicked Floquet Ising model quenched from Nèel State. QPU time = 7m 48s. Highest 2Q depth = 200

Results(L=100)



Parameters	Values
L(num_qubits)	100
Jz	π/2
hx (θ/2)	π/2
hz	π/2
Error Mitigation	ZNE (1, 1.2, 1.4) + DD+ MM + Twirling
Optimisation level	1

Figure.(Clifford vs QPU) Imbalance vs time for N=100 Kicked Floquet Ising model quenched from Nèel State. QPU time = 8m 53s. Highest 2Q depth = 200

Results(L=100)

Floquet Step	Imbalance(hardware)	Imbalance (Clifford)	Absolute Difference
5	-1.139428	-1	0.139428
10	1.191102	1	0.191102
15	-1.139535	-1	0.139535
20	1.130892	1	0.130892
25	-1.148508	-1	0.148508
30	1.094031	1	0.094031
35	-1.032225	-1	0.032225
40	0.981044	1	0.018956
45	-0.877113	-1	0.122887
50	0.805764	1	0.194236

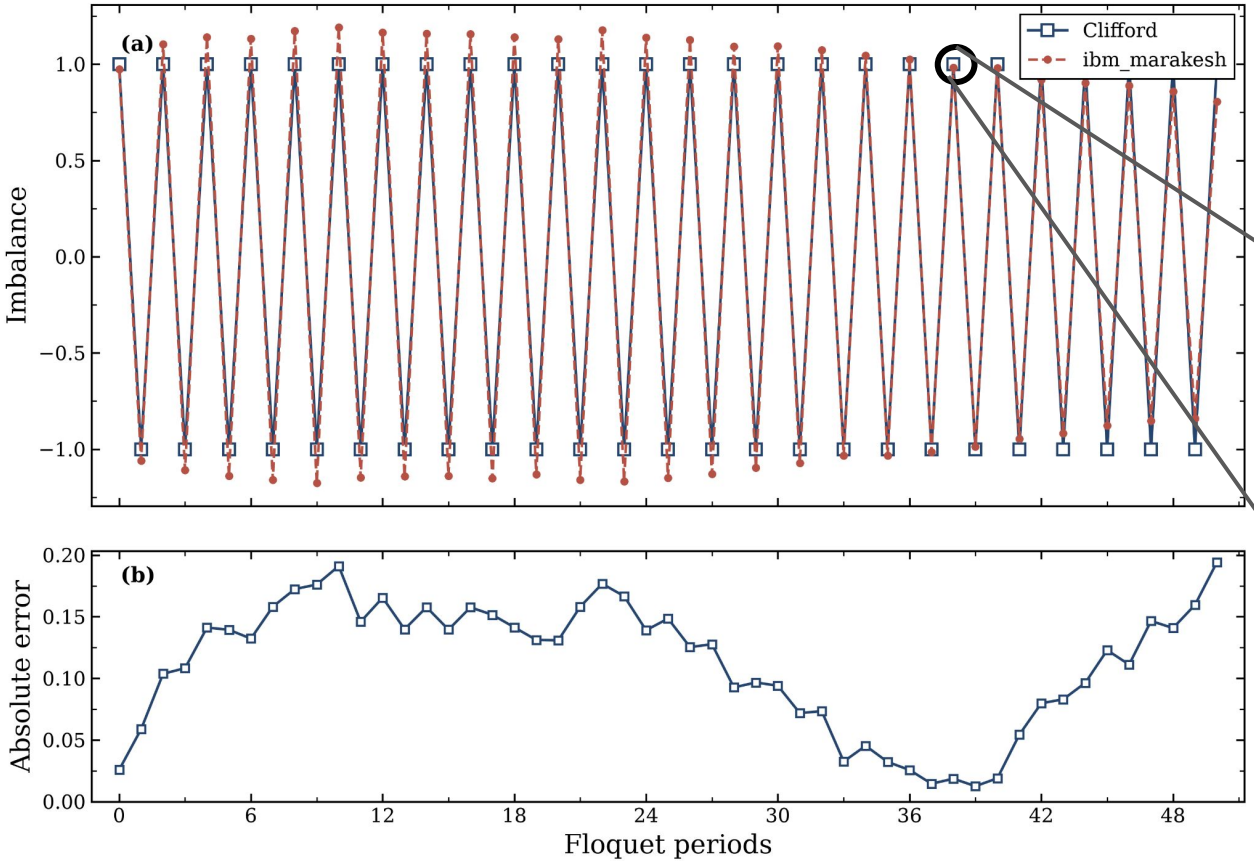
Table.(QPU) Imbalance vs time for N=100 Kicked Floquet Ising model quenched from Nèel State. This table shows few data points and compare it with reference and find absolute difference

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Results(L=100 Floquet Period = 40)



Metric	Clifford	Non-Clifford
2Q depth	160	160
Total depth	483	723
CZ gates	7,920	7,920
SX gates	7,842	17,929
X gates	4,138	50
RZ gates	15,812	18,029

Figure.(Clifford vs QPU) Imbalance vs time for N=100 Kicked Floquet Ising model quenched from Nèel State. QPU time = 8m 53s. Highest 2Q depth = 200

Conclusion and Future direction

- **Hypothesis**:- Non-Clifford Floquet evolution may remain experimentally accessible up to 40–50 periods
- Compare outcomes of non-clifford circuits with best classical methods like Pauli propagation and tensor network methods
- Apply different error mitigation strategies like ODR, PNA, etc.
- We can also simulate in 2D in square lattice with latest ibm nighthawk processor

Thank You